Kathmandu University

Department of Computer Science and Engineering

Dhulikhel, Kavre

A Project Report on

"Senti-Surfer : A Bilingual Approach to Sentiment Analysis for Social Media"

[Code No: COMP 207]

(For partial fulfillment of 4th Semester in Computer Science)

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Bona fide Certificate

This project work on

“Senti-Surfer : A Bilingual Approach to Sentiment Analysis for Social Media”

is the bonafide work of

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Abstract

With internet access being more common to people, social media has seen an explosive growth in the last few years. With a user base of 4.74 billion and counting, an overwhelming volume of data is being generated every single day. This data also includes the opinions and sentiments expressed by users on various topics in multiple languages. This project combines the power of web scraping, machine learning, sentiment analysis, a subfield of Natural Language Processing (NLP) and React to create an innovative web extension that provides users with a way to gain meaningful insights from social media posts and comments. This insight can be in the form of the ratio of positive to negative feedback, a trend analysis with data visualization depicting how the sentiment of the users about a particular topic is changing over time, or the response of people across different countries. This data can assist in a wide number of things ranging from improved decision making in business and politics to better disaster response. Furthermore, generated data is also stored in a database for some duration so that expensive computation will not have to be performed each time multiple users request the analysis of the same topic.

Keywords: Web scraping, Machine learning, Natural language processing, Sentiment analysis, React, Data visualization

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Acronyms/Abbreviations

API	Application Programming Interface
EwNp	English with Nepali phonetics
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
RegEx	Regular Expressions
RoBERTa	A Robustly Optimized BERT Pretraining Approach
URL	Uniform Resource Locator

Chapter 1: Introduction

1.1 Background

The rapid growth of social media throughout recent years has allowed for everyone to express their opinions on various topics ranging from products, movies and sports to political issues, religion and climate change. As of October 2022, It was found that social media users grew by 190 million over the past year, to reach a total of 4.74 billion. That global total has increased by 4.2 percent over the past 12 months, and social media users now equate to 59.3 percent of the world's total population (Kemp S, 2023). A wide amount of content is generated by these users every single day on social media platforms like YouTube, Twitter, Reddit, etc. With this, many opinions and sentiments are also generated. However, with such an overwhelming volume of data being generated, it can be difficult to make sense of it all and gain meaningful insights. This is where our project comes in.

In this project, we have developed a web extension that uses machine learning models trained on large datasets to delve into YouTube comments and accurately uncover the sentiments and opinions of the people on a video. This is achieved through sentiment analysis, a natural language processing (NLP) technique that identifies important patterns of information and features from a large text corpus. It analyzes thought, attitude, views, opinions, beliefs, comments, requests, questions, and preferences expressed by an author based on emotion (Lamba, M., & Madhusudhan M, 2021). The AI component of the web extension coupled with a user-friendly interface allows users to easily gain insights about the video. It can be used to identify whether a video is good or bad before the user commits to watching it. Furthermore, The web extension also provides interactive data visualizations including trend analysis showing how opinions and sentiments are changing over time, choropleth map to observe the average sentiment score of people across countries, world cloud to

display commonly used words and Radar chart to display the response of users across various domains like fear, joy, anger, disgust, sadness, and surprise.

In the context of Nepal, with current tools, the sentiments of the users can only be analyzed in one of two languages, English or Nepali. However, In practice, It is often observed that most comments are written in English but with the use of Nepali phonetics or expressions. For instance, a product review stating “dherai mahango vayo”. We can refer to such content as English with Nepali phonetics (EwNp). Hence we have built a tool for the collective sentiment analysis of such feeds containing 3 components, English, Nepali and EwNp.

1.2 Objectives

- To make sentiment analysis technology accessible to a wider audience through a user-friendly and web extension.
- To provide a tool for making more informed decisions in various domains, such as business, politics, disaster response, and more, by analyzing the emotions and opinions expressed by the users on social media platforms.
- To develop sentiment analysis tool for English, Nepali, EwNp text for social media platforms.

1.3 Motivation and Significance

The immense volume of data being generated on social media provides a great opportunity but gaining insights from this data can be a challenge. Current tools are limited by their complexity, expensive subscription, restriction to a single language or their use in very specific domains. The motivation behind this project is to provide people with a user-friendly, accessible, and comprehensive solution for analyzing bilingual social media content. For instance, with the removal of the youtube dislike counter, it has been increasingly difficult to gauge the quality of the videos, without watching it fully. Our tool can, in a way, return the dislike counter as a ratio of comments with positive sentiment and negative sentiment. In the context of Nepal, the comment can exist in English, Nepali or EwNp. Hence, all these forms of content need to be analyzed. Furthermore, It can be used to depict how the sentiments of the users over a certain topic or a person is changing over time so they make more informed decisions. It can also be used in more important matters like disaster response. It can be used to identify the most pressing needs of affected individuals. For instance, if most social media posts express frustration about slow response time, then the concerned authorities can prioritize it for better risk management. Sentiment analysis can also be widely used in the stock market to track public opinion of individual stocks.

Chapter 2: Related Works/ Existing Works

2.1 Talkwalker

Talkwalker (Talkwalker Inc., 2009) offers a tool called Quick Search that enables users to track mentions of particular keywords, hashtags, and businesses on social networking platforms. Quick Search users have access to sentiment analysis, historical data in real-time as well as into online conversations. Talkwalker offers a range of visualization tools, such as graphs and charts, to help users understand the results of their search. It is a useful tool for monitoring brand mentions, monitoring online reputation. It is, however , a paid service for business professionals.

Fig 2.1.1: Talkwalker.com

2.2 Lexalytics

Lexalytics (Lexalytics, 2003) is a text analysis tool that can be used to identify whether a piece of text is negative, positive, or neutral . It uses natural language processing (NLP) and machine learning algorithms to help companies analyze large amounts of unstructured text data, such as customer reviews, social media posts, and survey responses.It also contains a variety of visualization tools, such as graphs and charts, to assist users interpret the results of their search and share their findings with others.

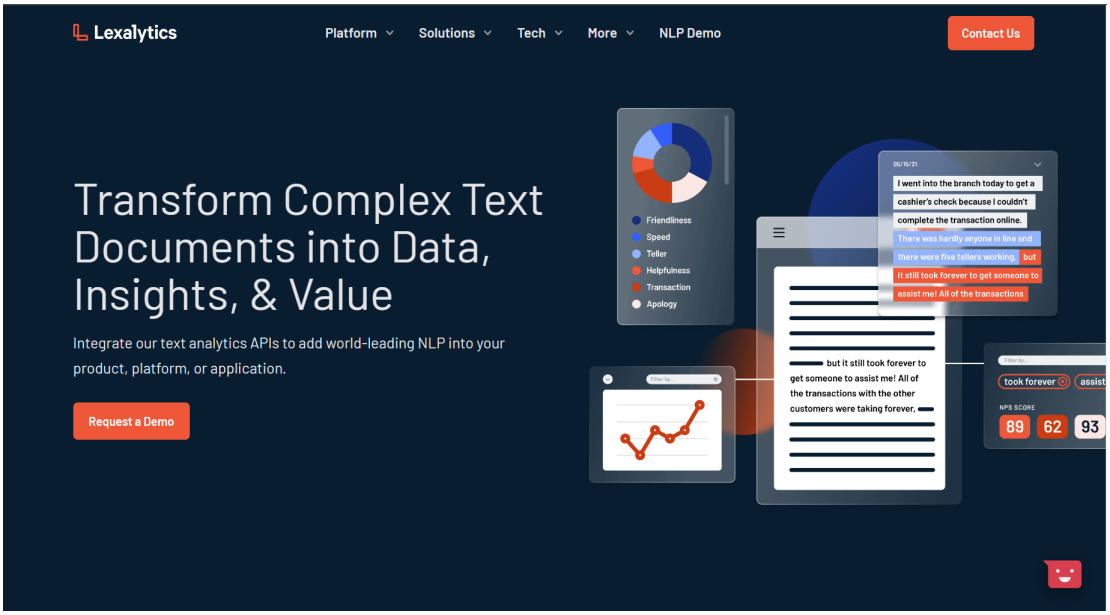

Fig 2.2.1: Lexalytics.com

2.3 Brand24

Brand24 (Brand24, 2011) is a sentiment analysis app that helps businesses in keeping track of brand mentions on social media. Businesses are able to quickly and effectively respond to consumer demands and concerns thanks to the real-time insights it provides into customer sentiment, opinion, and behavior. This app uses natural language processing and machine learning algorithms to determine the emotional tone of social media posts, giving businesses an overview of consumer perceptions of their brand.

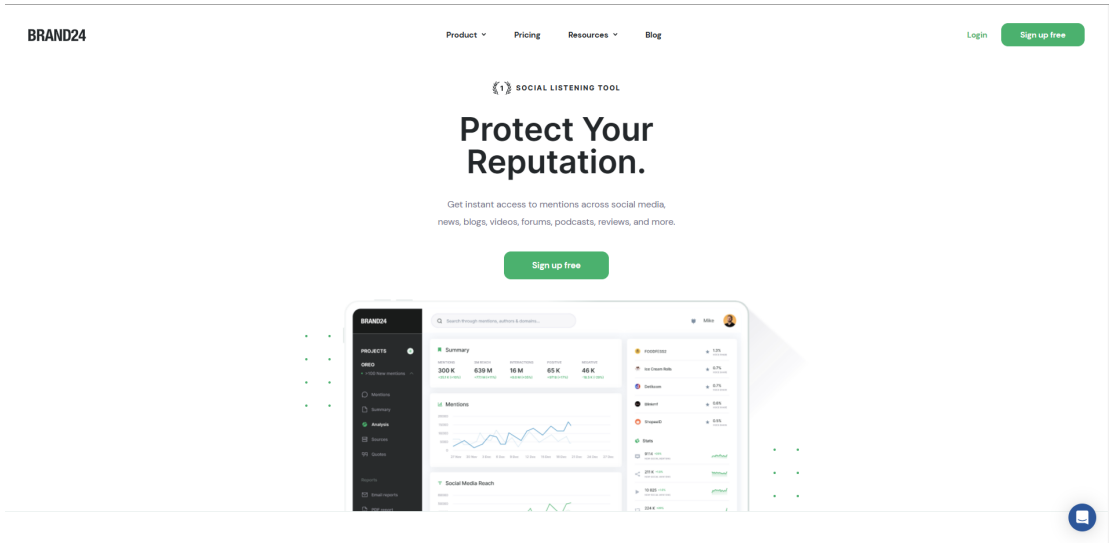

Fig 2.3.1: Brand24.com

Chapter 3: Design and Implementation

3.1 System Requirement Specifications

3.1.1 Software Specifications

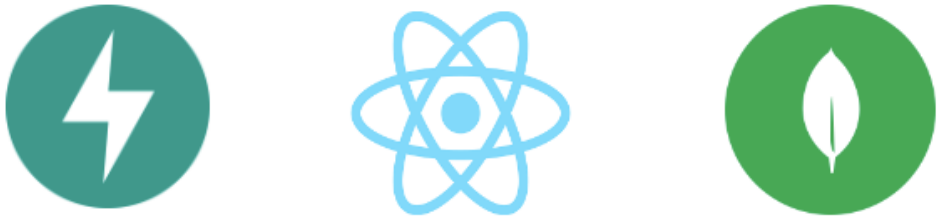

Fig 3.1.1 : FARM Stack

This project is made using the FARM stack. Farm Stack is a modern web development framework that combines FastAPI, React, and MongoDB.

3.1.1.1 Front End Tools

React.js: React is a frontend JavaScript library for building fast and interactive UI components. React can be used as a base in the development of single page applications. React is one of the most popular JavaScript libraries used for building UI.

Tailwind CSS: Tailwind CSS is a popular CSS framework for building user interfaces. It provides a set of pre-designed CSS classes that can be used to quickly style web pages without having to write custom CSS code. It helps in ensuring a consistent design system throughout the app.

Chart.js: Chart.js is a powerful JavaScript library for creating interactive and visually appealing charts and graphs on the web. With Chart.js, developers can customize colors, labels, tooltips, and animations to create stunning data visualizations like barchart, radarchart, pie chart and many more that enhance data understanding and storytelling.

3.1.1.2 Back End Tools

FastAPI: FastAPI is a modern, high-performance web framework for building APIs with Python. It offers incredible speed and efficiency by leveraging type annotations and asynchronous programming, making it an excellent choice for building scalable and robust web applications.

MongoDB: MongoDB is a popular NoSQL database known for its flexibility and scalability. It offers a document-oriented data model, allowing developers to store and retrieve data in a JSON-like format. With its distributed architecture and automatic sharding capabilities, MongoDB can handle large volumes of data and support high-performance applications.

Transformers: Transformers is a cutting-edge natural language processing (NLP) technology that revolutionized the field. Transformers have enabled breakthroughs in various NLP tasks, including machine translation, sentiment analysis, question answering, and more. This project uses 2 models based on RoBERTa. emotion-english-distilroberta-base (j-hartmann, 2022) is used for emotion classification (fear, joy, anger, disgust, sadness, and surprise) and twitter-roberta-base-sentiment (cardiffnlp, 2020) is used for sentiment classification (positive, negative, neutral).

3.1.2 Hardware Specifications

This project is a web extension so a modern browser (Google Chrome, Microsoft Edge, etc.) should be installed.

3.2 Project Workflow

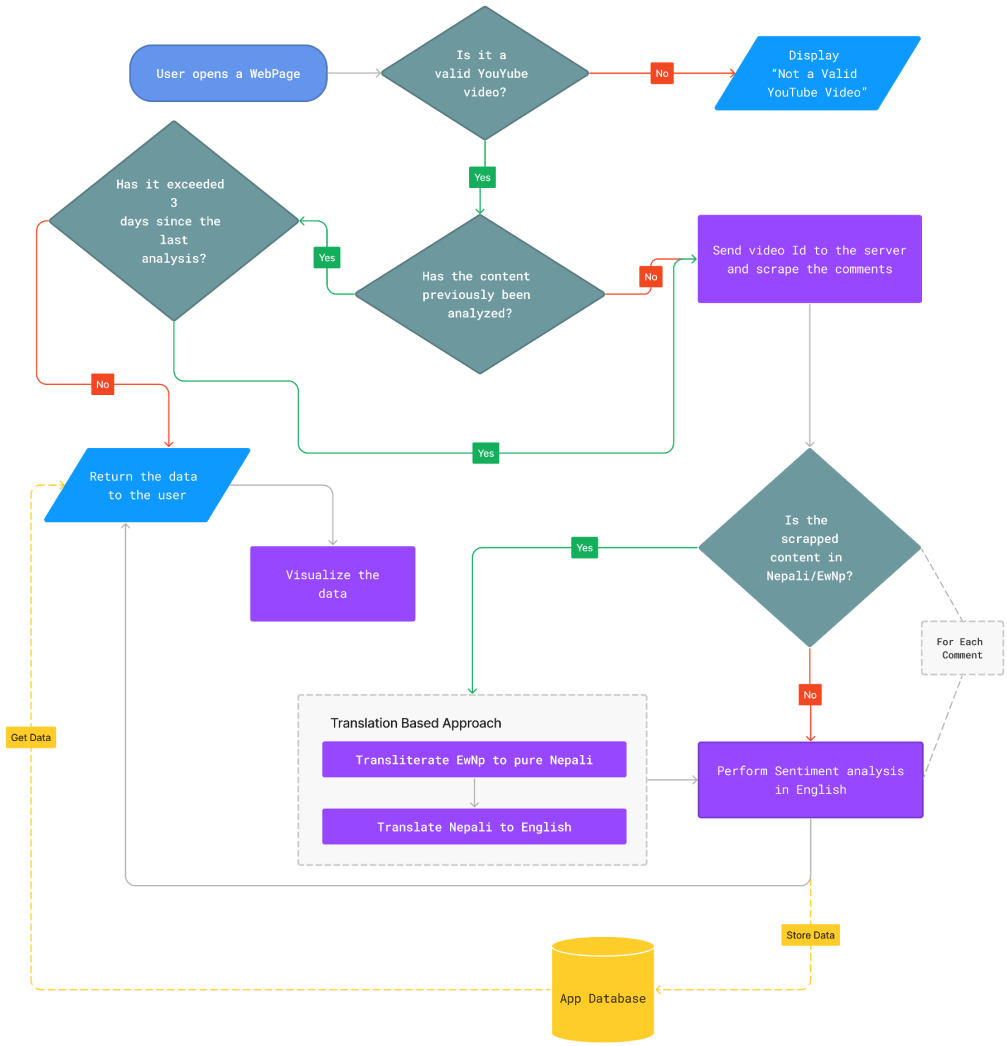

Fig 3.2.1 : Project Workflow

Upon opening the extension, the user is displayed whether they can analyze the contents of the website they are in. For the first phase of this project, only comments from a YouTube video can be analyzed. Hence, the analyze option is only available if the user is in a valid YouTube video URL. This is checked with the help of RegEx (Regular Expressions).

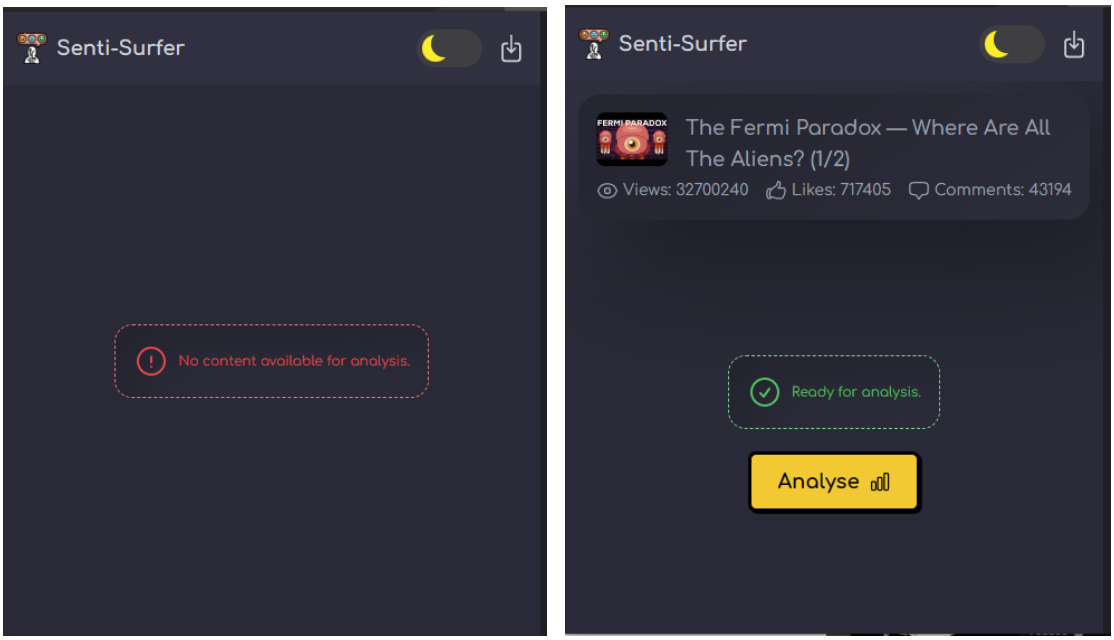

Fig 3.2.2 : Invalid and Valid URLs

If the user is in a valid YouTube video, Video statistics is displayed along with an analyze button. On pressing the analyze button, the videoId is sent to the server which first checks if the video has been analyzed within the past 3 days. If it is, then the analyzed data is sent back to the user. This is done to minimize computation cost and save time since analyzing a video can take up to a minute. If it is not, then the comments and location data of the commenter is scrapped by the server. Comments having EwNp are transliterated to pure Nepali which is then translated to English and this data is fed into our model to generate predictions about sentiment of each comment. The analyzed results are then saved to the database for future use and returned to the user.

```
_id: ObjectId('646195cf1ab22a4d56de22ee')
videoId: "d6yAnHb6EKA"
▸ data: Array
analysed_date: "2023-05-15"
```

```
data: Array
▾ 0: Object
    text: ". Free Wallpapers: https://bit.ly/Pinehill_Wallpapers"
    author: "Melody Park♬"
    date: "2023-05-02T15:34:54Z"
    positive: 0.2494739145040512
    negative: 0.02119551971554756
    neutral: 0.8851179480552673
    compound: 61.41391935674372
    sentiment: "neutral"
    channelId: "UCTxRAUXPEnXRyFuz2F27JBw"
    anger: 0.0026168476324528456
    disgust: 0.0011369576677680016
    fear: 0.0010177802760154009
    joy: 0.05506652593612671
    sadness: 0.008452675305306911
    surprise: 0.04659130796790123
    location: "USA "
▸ 1: Object
▸ 2: Object
```


Fig 3.2.3 : Data Formats

The analyzed result is stored in JSON format as key value pairs. The videoId key contains the unique Id of the analyzed video and the data key contains an array of objects. Each object represents a comment with attributes describing it. This data is used by the front end for visualizations.

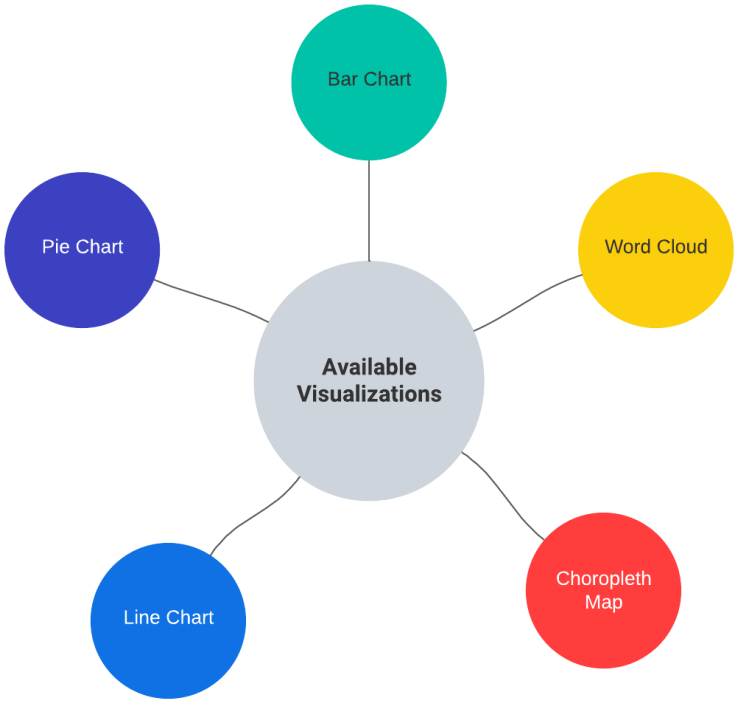

Fig 3.2.4 : Available Visualizations

A total of 6 data visualizations are rendered in the front end using the data returned by the backend. Bar Chart is used to display the number of positive, negative and neutral comments. Word Cloud is used to visualize the most commonly used words. Choropleth Map is used to display the average sentiment score from each country. Line Chart is used to show how the sentiment of the comments and emotions are evolving through time and Pie Chart is used to show the percentage of different emotions expressed in the comments.

Chapter 4: Discussion of the achievements

4.1 Features

4.1.1 Display the average sentiment score

The web extension determines the positive polarity score, negative polarity score and the neutral polarity score for each comment. Following that, a weighted compound score is determined for each comment which ranges from 0-100 which signifies the average polarity of a comment. The average of the weighted scores of the comment is calculated.

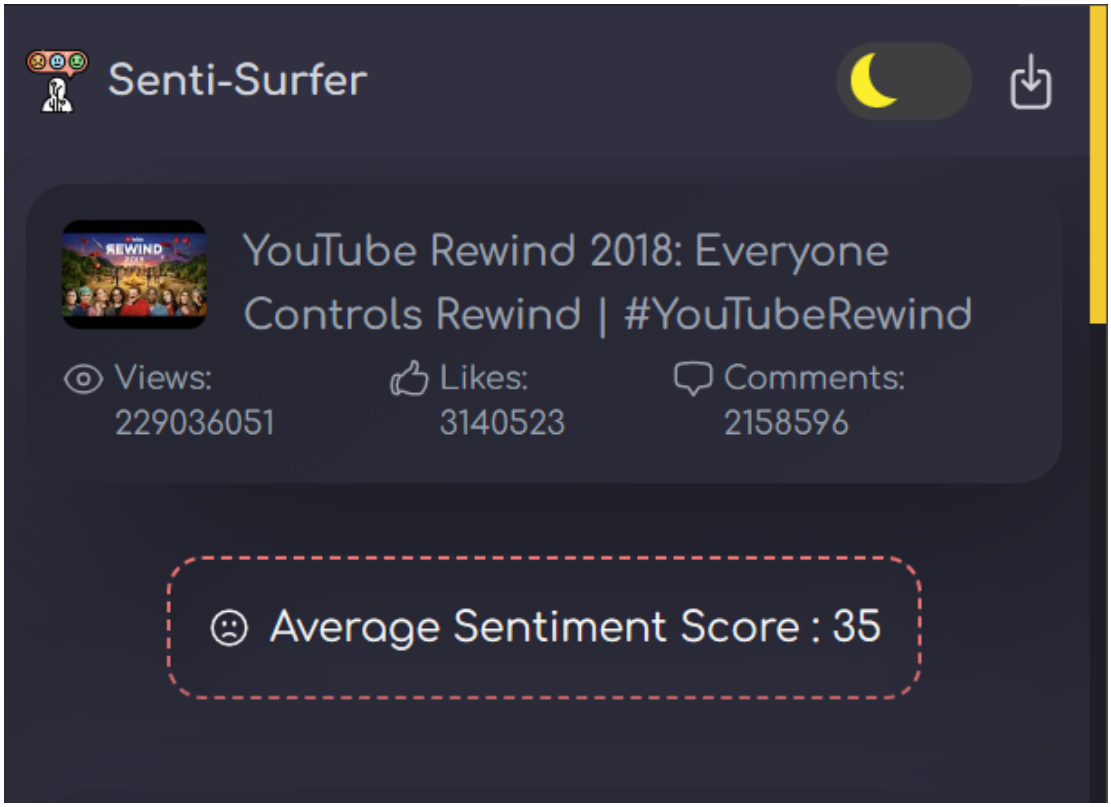

Fig 4.1.1.1 : Average Sentiment Scores

4.1.2 Display the number of positive, negative and neutral comments

The server does not just determine the sentiment polarity score of each comment, but it also determines if the comment is positive, negative or neutral. The total counts of each comment is calculated and this data is used to create a bar chart with counts on the y-axis and positive, negative and neutral labels on the x-axis.

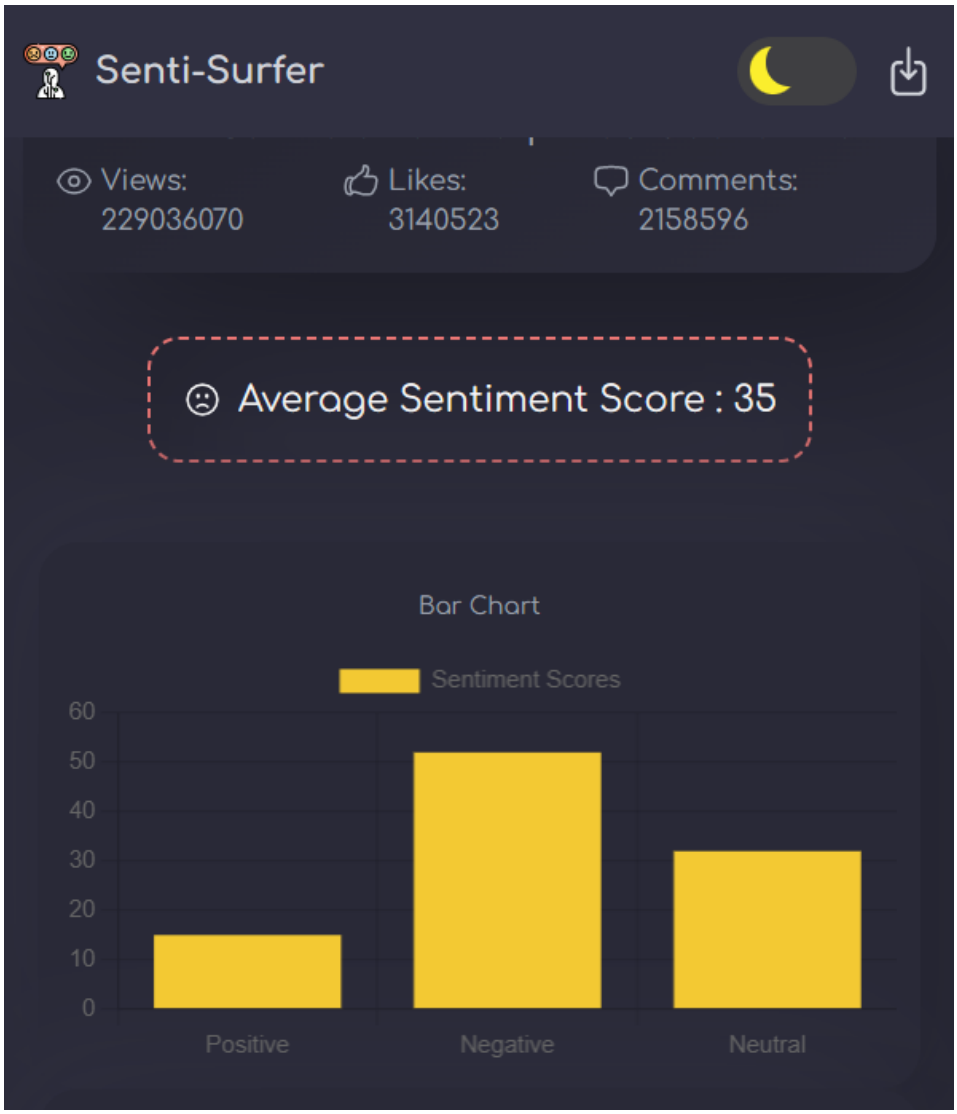

Fig 4.1.2.1 : Bar Chart

4.1.3 Display how the sentiment is evolving over time

The web extension also displays how the sentiment score is evolving over time with the help of a line chart. The average sentiment score across each date is calculated and this data is used to generate the line chart. Furthermore, it also displays the trend line for different emotions expressed by users.

Fig 4.1.3.1 : Trend Analysis

4.1.4 Display response of users from different countries

After the comments are fetched, the location data of the commenter, if available, is also determined with the help of web scraping. Following that, the average sentiment score from each country is also to create a choropleth map which depicts the sentiments of the commenters from different countries.

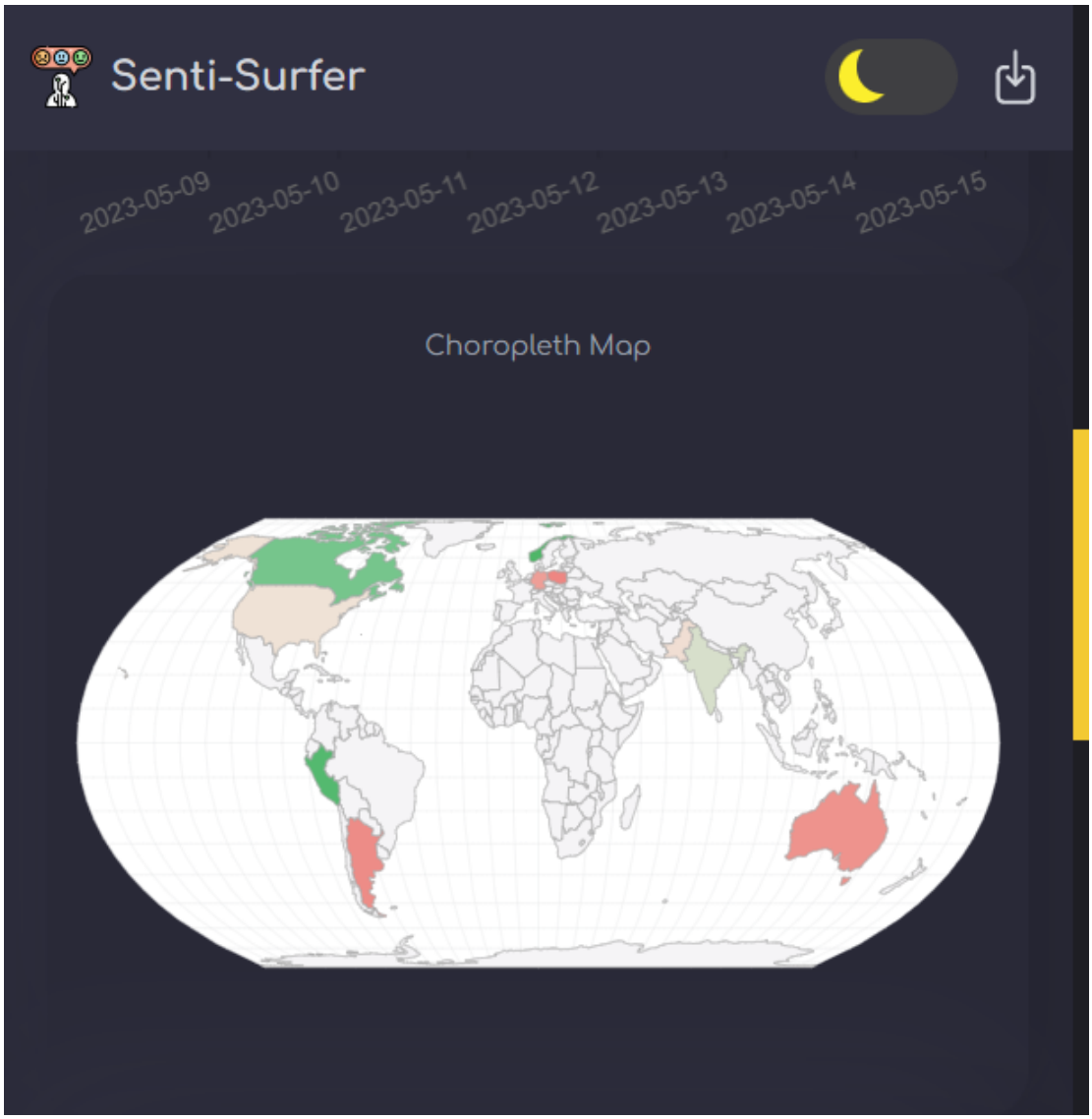

Fig 4.1.4.1 : Choropleth Map

4.1.5 Display the most common words used in the comments

The number of occurrences of each word in the entire comment corpus is calculated. Following that the stop words, i.e commonly used words like this, that, an, etc are removed. The most commonly used words are then extracted and a world cloud is rendered.

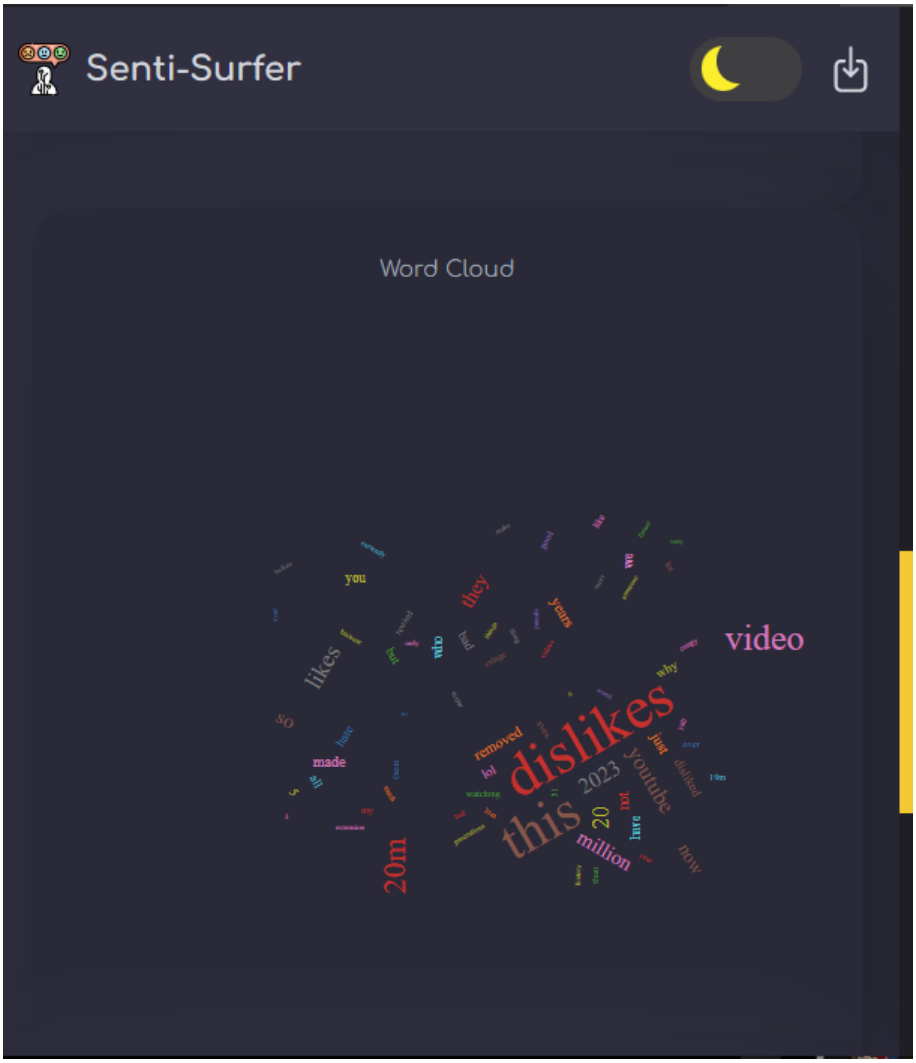

Fig 4.1.5.1 : Word Cloud

4.1.6 Display the emotion expressed by the comments

A comment can express various emotions like anger, disgust, fear, joy, neutral, sadness and surprise. This data is returned in the form of scores for each of this comment. The average score for each emotion across all the comments is calculated to display a pie chart.

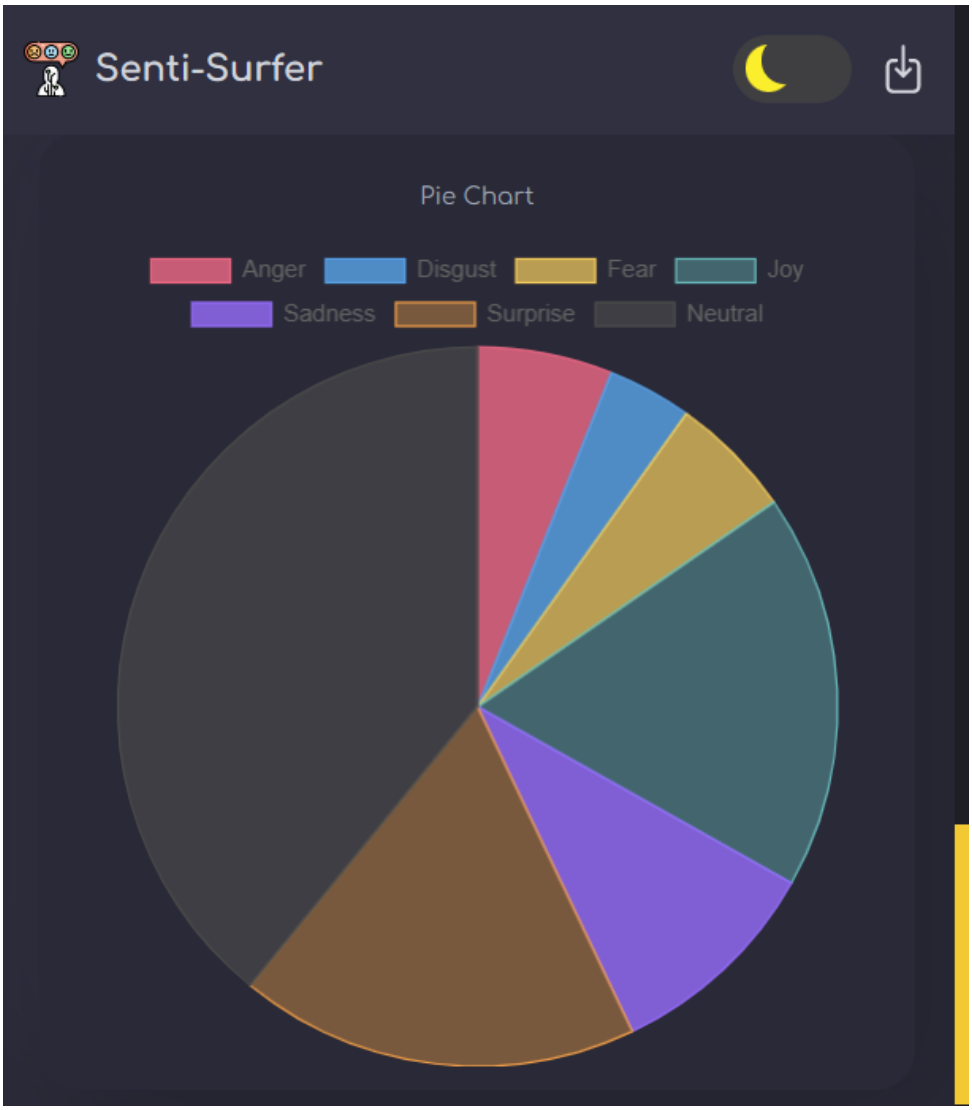

Fig 4.1.6.1 : Pie Chart

4.2 Challenges

1. Visualization of the JSON data was challenging using chart.js due to complex data structures, extraction and mapping the data to appropriate chart elements.
2. Understanding machine learning and neural networks required knowledge of fundamentals of machine learning algorithms and the basic structure and functioning of neural networks which was challenging.
3. Developing a Chrome extension was a challenging task which required careful planning and implementation.

4.3 Achievements and Contributions

1. Developed a sentiment analysis tool to analyze comments for YouTube.
2. Successfully integrated a pre-trained sentiment analysis model into the project.
3. Implemented data visualization features to help users better understand sentiment analysis results.
4. Provided a valuable tool for businesses and organizations to analyze customer feedback and improve their services/products.
5. Explored the potential of sentiment analysis for Nepali language and paved the way for future development in this area.

Chapter 5 : Conclusion and Recommendation

Leveraging the power of web scraping, machine learning and data visualization, this project provides a platform to take a peek into the minds of thousands of people and to discover the joy, anger, sorrow, fear and opinions of the world. It allows us not only to view a sentence as positive or negative, but to see beyond it and understand the underlying emotions and opinions. With the help of this web extension, we can gain a deeper understanding of the world around us, and use that knowledge to make more informed decisions.

5.1 Limitations

1. Nepali sentiment analysis currently relies on translations to English, which may not fully capture the nuances and context of the original language.
2. The sentiment analysis may not be able to accurately identify sarcasm or irony in comments.
3. The response from users in different countries may be limited by the availability and quality of data.
4. Scraping YouTube data and generating analysis can take some time, especially for large datasets.

5.1 Future Enhancements

1. Develop a more accurate and comprehensive sentiment analysis model that can handle Nepali language directly without the need for translation.
2. Improve the model to detect sarcasm and irony in sentences.
3. Include additional visualizations for specific user groups like age and gender.
4. Integrate with social media platforms to automatically collect and analyze comments in real-time.

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Annex

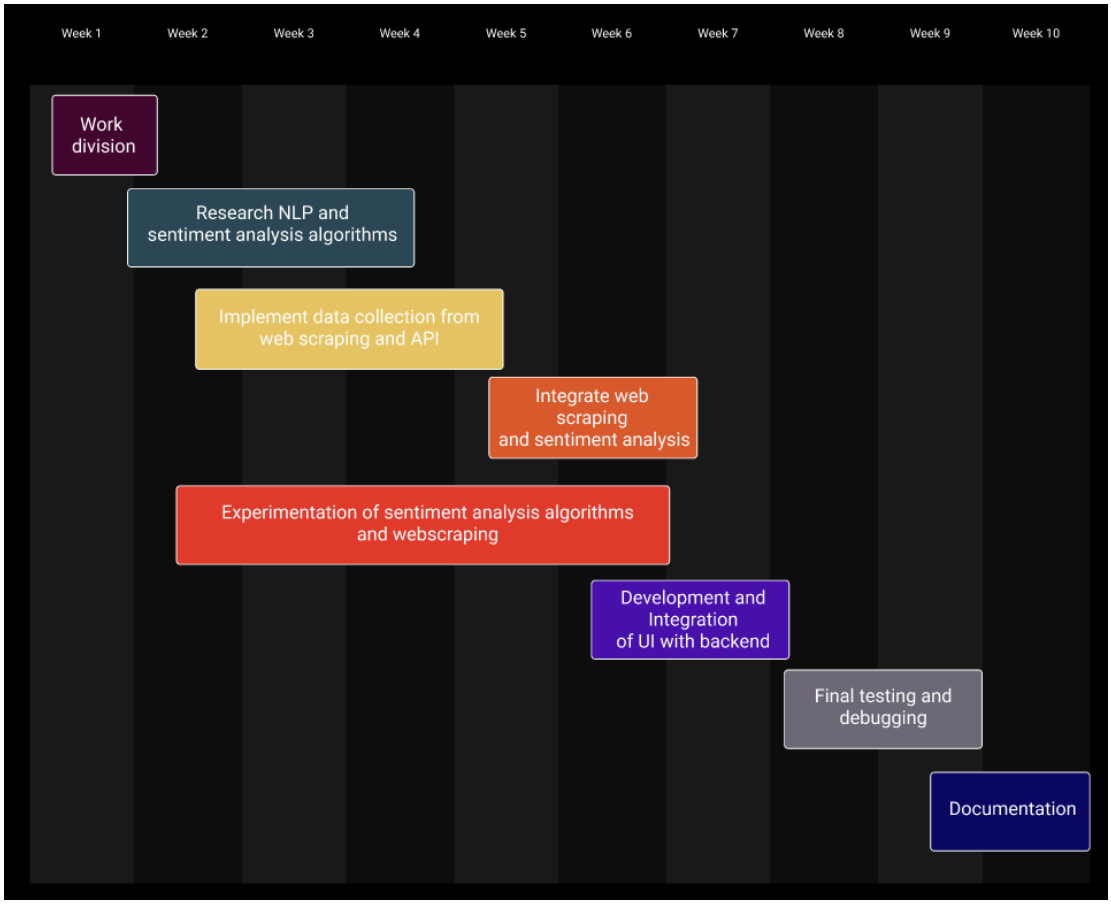

Fig A.1: Gantt Chart